

MSc AI Engineering

Self-indication of admission requirements and selection criteria

Application number:	3536719 (Uni-assist Number)
Surname/Family name:	Nagpal
First name/Given name:	Saksham
Date of birth:	17.07.2003
Bachelor's degree:	Bachelor of Engineering in Computer Science & Engineering
Awarding University (name and place):	Chitkara University, Punjab, India

Remember: filling out the form as well as possible will make the application process smoother and faster.

Admission requirements

Please list all courses in your degree program that you have successfully passed in the form below.

Course title ¹	Institution ²	Grade	Credit hours
Introduction to Web Technologies	Chitkara University, Punjab, India	O	4
Problem Solving Using Python Programming	Chitkara University, Punjab, India	O	5
Next Gen Technologies	Chitkara University, Punjab, India	O	2
Cyber Security	Chitkara University, Punjab, India	O	2
Basics of Electronics Engineering	Chitkara University, Punjab, India	O	4
Basics of Electronics Engineering Lab	Chitkara University, Punjab, India	O	1
Introduction to Emerging Technologies	Chitkara University, Punjab, India	B+	2
Calculus and Statistical Analysis	Chitkara University, Punjab, India	O	5
Differential Equations and Transformations	Chitkara University, Punjab, India	O	5
Object Oriented Programming Using C++	Chitkara University, Punjab, India	O	4
Advanced Web Development	Chitkara University, Punjab, India	O	4
Source Code Management	Chitkara University, Punjab, India	O	2
Operating System Fundamentals with Linux	Chitkara University, Punjab, India	O	4
Modern and Computational Physics Lab	Chitkara University, Punjab, India	C	1
Modern and Computational Physics	Chitkara University, Punjab, India	O	4
Human Values and Professional Ethics	Chitkara University, Punjab, India	Satisfactory	-
Discrete Structures	Chitkara University, Punjab, India	A+	4
Data Structures	Chitkara University, Punjab, India	A	5
Database Management System	Chitkara University, Punjab, India	A+	4
Theory of Computation	Chitkara University, Punjab, India	A+	4
Cyber Science and Ethical Hacking	Chitkara University, Punjab, India	A+	2
Artificial Intelligence	Chitkara University, Punjab, India	O	2
Engineering Exploration (Roadmap to Blockchain)	Chitkara University, Punjab, India	O	2
Disaster Management	Chitkara University, Punjab, India	Satisfactory	-
Core Java	Chitkara University, Punjab, India	A	5
Cloud Computing Architecture	Chitkara University, Punjab, India	A+	4
Problem Solving Using C	Chitkara University, Punjab, India	B+	5
Linux System Administration	Chitkara University, Punjab, India	A+	4
Introduction to Computer Networks	Chitkara University, Punjab, India	A	4
Computer Organization and Architecture	Chitkara University, Punjab, India	A	3
Digital Marketing	Chitkara University, Punjab, India	O	4
Algorithm Design & Implementation	Chitkara University, Punjab, India	C	4
Programming Abstractions	Chitkara University, Punjab, India	B+	4
Front-End Engineering	Chitkara University, Punjab, India	A+	4

Back-End Engineering	Chitkara University, Punjab, India	A+	4
Professional Practices-System Design	Chitkara University, Punjab, India	B+	2
Numerical Ability and Logical Reasoning	Chitkara University, Punjab, India	O	4
Vital Employability Skills	Chitkara University, Punjab, India	O	2
Google UX Design Professional Certificate	Chitkara University, Punjab, India	O	2
Object Oriented Software Engineering	Chitkara University, Punjab, India	A	4
Cloud Computing	Chitkara University, Punjab, India	B+	4
App Prototyping Using Swift Fundamentals	Chitkara University, Punjab, India	A	4
App Development Skills	Chitkara University, Punjab, India	A+	4
Industry Immersion Program	Chitkara University, Punjab, India	A+	2
Environmental Sciences	Chitkara University, Punjab, India	A+	2
Algorithms and Programming Paradigms	Chitkara University, Punjab, India	O	4
Advanced Web Design Frameworks & Development	Chitkara University, Punjab, India	O	4
Exploratory Project	Chitkara University, Punjab, India	B+	4
Industry Oriented Hands-On Experience	Chitkara University, Punjab, India	O	12
Art of the Job Interview	Chitkara University, Punjab, India	Satisfactory	-

¹Please indicate the title of the course, e.g. “Introduction to Computer Science”, or similarly. Avoid abbreviations.

²Please indicate the institution that awarded the certificate for the course. ³Please indicate the – unweighted - number of credits (in units at your University) you received for the successful completion of the course. Do not worry about converting your credits into ECTS – we will sort it out. If you have completed additional courses at a different university (e.g. during an exchange programme or as part of another degree such as an additional master’s), please, add them to the list as well.

Required credits and module catalogue of your Bachelor’s degree

Please state how many credits (or equivalent units) are necessary to complete your Bachelor’s degree, independently whether you already have acquired them or are going to:

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Please also provide a **link to the module catalogue/syllabus** of your Bachelor’s degree programme. If you are unable to provide a link, please upload the module catalogue together with the other files.

https://drive.google.com/drive/folders/1EfCQEAK9q0ESTdZlSGuX7U2vve2seCWq?usp=drive_link

Individual Contribution

Was your dissertation/thesis/final year project an individual project or a group project? In case you have been collaborating with others, please indicate below your individual contribution. Mention not only the technical, but also methodological or project-related aspects

This was an individual project supervised by Mrs. Preeti Saini (Assistant Professor, Chitkara University). I independently conceived and developed Airsecure — a dual-purpose AI-powered web application combining real-time restricted object detection with face recognition-based attendance management. Technically, I integrated YOLOv10n for real-time object detection via webcam feed, implemented face recognition using OpenCV and Dlib for automated attendance tracking, and built the full backend using Python, Flask, and SQLAlchemy with RESTful APIs. I designed and developed both the teacher and student dashboards using HTML, CSS, and JavaScript, including attendance analytics, QR code generation, and text-to-speech alert integration. Methodologically, I independently defined the problem scope, designed the system architecture, selected and integrated AI models, and structured a phased development plan from initiation through deployment. All code, documentation, and testing were completed solely by me.